

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Authentication	Registration via email & password Login with role-based access (Planner / Analyst / Admin) Password reset via email
FR-2	Data Ingestion	Upload state-wise consumption CSV files Validate & flag missing or duplicate entries Store cleaned records in the central database
FR-3	Dashboard & Visualization	Real-time regional consumption dashboard Interactive geographic heatmap (lat/long) Year-over-year trend comparison charts
FR-4	Demand Forecasting	Generate short-term (7-day) demand forecasts Seasonal trend and anomaly detection
FR-5	Alerts & Notifications	Peak-load threshold alerts Email and in-app notifications
FR-6	Reporting & Export	Export dashboards/forecasts as PDF or Excel Scheduled report emails to stakeholders

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboard is simple and intuitive enough to use after a single 15-minute walkthrough, with no specialized training required.
NFR-2	Security	Role-based access control, encrypted credentials at rest, HTTPS/TLS for all traffic, and audit logging of data exports.
NFR-3	Reliability	The dashboard and alerting service maintain 99.5% uptime during business hours (6 AM–10 PM IST).
NFR-4	Performance	Dashboard queries return results in under 3 seconds, even for a 5-year date range across all states.
NFR-5	Availability	Multi-region hosting with automatic failover so a regional cloud outage does not take the platform offline.
NFR-6	Scalability	Architecture supports adding new states, regions, or data sources without redesigning the ingestion or dashboard layers.